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$$f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto x^2 - 1 \quad (a = 2)$$

$$\varepsilon > 0$$

$$\delta = \min\left(1, \frac{\varepsilon}{5}\right)$$

$$|x - 2| < \delta$$

$$|f(x) - f(2)| = |x^2 - 1 - (4 - 1)|$$

$$= |x^2 - 4|$$

$$= |x - 2||x + 2|$$

$$|x - 2| < 1 \Rightarrow 1 < x < 3$$

$$|x + 2| \leq 5$$

$$|f(x) - f(2)| < \delta \cdot 5$$

$$\leq \frac{\varepsilon}{5} \cdot 5$$

$$= \varepsilon$$

$$f \text{ continu in } a = 2$$

References